

1. Administrative & Study Identification

Full Study Title: A Phase II, Multi-part, Randomized, Open-label, Assessor-blinded, Active-controlled, Multicenter Study to Evaluate the Efficacy and Safety of Rapcabtagene Autoleucel Versus Rituximab Treatment in Participants With Severe Refractory Diffuse Cutaneous Systemic Sclerosis **Short Title/Acronym:** Phase 2 Study Evaluating Rapcabtagene Autoleucel in Participants With Diffuse Cutaneous Systemic Sclerosis **Protocol Number:** CYTB323K12201 (Other ID: 2023-510380-34-00) **NCT Number:** NCT06655896 **Posting ID:** 561005131862448168 **Sponsor/Company Name:** Novartis Pharmaceuticals **Investigational Product:** Rapcabtagene Autoleucel (YTB323) / Active Control: Rituximab (Trade Names: Rituxan, Riabni, Ruxience, Truxima, Rituxan Hycela) **Phase of Development:** Phase II **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and tolerability of rapcabtagene autoleucel (administered once following lymphodepletion) in participants with severe refractory diffuse cutaneous systemic sclerosis (dcSSc) relative to rituximab. **Secondary Objectives:** To evaluate functional ability and disability changes (via HAQ-DI), pulmonary preservation (via FVC), skin fibrosis reduction (via mRSS), and the long-term safety profile of anti-CD19 CAR-T therapy.

2.2 Background & Rationale To address the critical unmet medical need in patients with severe refractory diffuse cutaneous systemic sclerosis (dcSSc). Standard B-cell depleting therapies (like rituximab) have shown some efficacy, but severe, progressive cases require more potent disease-modifying options. Rapcabtagene autoleucel, a next-generation CAR-T therapy with preserved T-cell stemness and a rapid manufacturing process, may offer deeper and more sustained depletion of pathogenic B-cells, potentially halting or reversing severe skin and lung fibrosis.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Rituximab) **Blinding:** Open-label, Assessor-blinded (to prevent bias in clinical scores like mRSS) **Randomization:** Randomized (Includes a Lead-in Cohort receiving CAR-T, followed by a Randomized Cohort receiving CAR-T or rituximab)

3.2 Planned Enrollment & Duration Target N: 86 Number of Sites: Multicenter (United States + 17 other locations) **Estimated Study Start:** 28-Oct-2024 **Estimated Primary Completion:** 29-Aug-2032 (Includes 5-year follow-up)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female, 18 to 70 years of age. 2. Participant must fulfill the 2013 American College of Rheumatology/European League Against Rheumatism classification criteria for systemic sclerosis and meet the diffuse cutaneous SSc (dcSSc) subset classification according to LeRoy. 3. Disease onset from the first non-Raynaud symptoms attributable to SSc (e.g., puffy hands, scleroderma, digital ulcers, arthralgia, dyspnea) within 7 years prior to the Screening visit. 4. Severe, progressive systemic sclerosis disease defined by at least one of the following: Progressive systemic sclerosis-associated interstitial lung disease, severe progressive systemic sclerosis skin disease, or clinically significant systemic sclerosis-associated cardiac involvement at Screening.

4.2 Exclusion Criteria 1. Any condition during Screening that could prevent a complete washout of medications as required per protocol or could otherwise make the participant ineligible for anti-CD19 CAR-T therapy and further participation in the study, as judged by the Investigator. 2. Participants with history of hypersensitivity to excipients in rapcabtagene autoleucel or to rituximab. 3. Any participant for whom treatment with rituximab is clinically inappropriate in the opinion of the investigator. 4. Any medical conditions that are not related to SSc that, in the opinion of the Investigator, would jeopardize the ability of the participant to tolerate lymphodepletion and anti-CD19 CAR-T cell therapy. 5. Rheumatic disease other than dcSSc, (except secondary Sjogren's syndrome or scleroderma myopathy), including limited cutaneous systemic sclerosis (lcSSc) or sine scleroderma at Screening. 6. Participants with pre-existing pulmonary hypertension. 7. Significant renal pathology at Screening. 8. Participants with uncontrolled stage II hypertension at Screening. 9. Vaccination with live attenuated vaccines within 6 weeks prior to randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Investigational Arm:** Lymphodepleting therapy with fludarabine (adjusted for renal function) and cyclophosphamide daily for 3 days, followed by a single intravenous infusion of rapcabtagene autoleucel. * **Active Control Arm:** Rituximab standard dosing. (Note: Patients in the rituximab arm whose disease is not fully controlled may crossover to receive rapcabtagene autoleucel if eligible per protocol). **Route of Administration:** Intravenous (IV) **Duration of Treatment:**

Single infusion for CAR-T, with long-term follow-up span of 5 years (and subsequent rollover to a separate LTFU protocol).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and symptom-management medications that do not interfere with T-cell or B-cell dynamics. **Prohibited:** Live attenuated vaccines within 6 weeks prior to randomization. Any medication preventing a complete protocol-required washout prior to lymphodepletion.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -45 to -14	Informed Consent, 2013 ACR/EULAR criteria verification, Medication Washout, Baseline Labs
Baseline	Day -5 to 0	Lymphodepletion (Fludarabine + Cyclophosphamide x3 days), Randomization
Infusion	Day 1	Rapcabtagene Autoleucel single IV infusion (or Rituximab initiation)
Interim	Week 52	Primary Efficacy Assessment (rCRISS50 components), Safety Labs
End of Study	Year 5 (Week 260)	Final Vital Signs, Exit Interview, Rollover to separate LTFU protocol

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Achievement of a treatment response as per the Revised Composite Response Index in Systemic Sclerosis 50 (rCRISS50) definition at Week 52. **Measurement Method:** To demonstrate superiority of rapcabtagene autoleucel compared to rituximab, this composite response is assessed across 5 domains: (1) modified Rodnan Skin Score (mRSS), (2) Health Assessment Questionnaire Disability Index (HAQ-DI), (3) patient global assessment (PGA), (4) physician global assessment (PhGA), and (5) percent-predicted forced vital capacity (FVC%).

7.2 Secondary & Exploratory Endpoints * Change from baseline in Forced Vital Capacity (FVC)% predicted at Week 52. * Change from baseline in modified Rodnan Skin Score (mRSS) at Week 52. * Change from baseline in Health Assessment Questionnaire Disability Index (HAQ-DI) at Week 52.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extensive monitoring for CAR-T specific toxicities (e.g., Cytokine Release Syndrome [CRS], neurotoxicity) and risks associated with lymphodepletion (cytopenias, infections). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline to the sponsor and applicable regulatory bodies. **Data Monitoring Committee (DMC):** Independent

external DMC utilized to review safety data during the Lead-in cohort prior to broader randomization.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy, Per-Protocol (PP), and Safety Set (all patients receiving any component of the study treatment). **Sample Size Justification:** Target N of 86 is powered to detect a statistically significant superiority in the proportion of participants achieving an rCRISS50 response at Week 52 in the rapcabtagene autoleucel arm versus the rituximab arm ($\alpha = 0.05$). **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) or Multiple Imputation for longitudinal primary and secondary endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict independent assessor-blinding protocols to ensure objective mRSS evaluations in an open-label setting. **Audit Trail:** Standard 21 CFR Part 11 compliant eCRF locking and query resolution.

11. Study Identifiers & Governance

Primary ID: CYTB323K12201 **Registry Number:** NCT06655896 **Posting ID:** 561005131862448168 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Phase 2 Study Evaluating Rapcabtagene Autoleucel in Participants With Diffuse Cutaneous Systemic Sclerosis **Official Regulatory Title:** A Phase II, Multi-part, Randomized, Open-label, Assessor-blinded, Active-controlled, Multicenter Study to Evaluate the Efficacy and Safety of Rapcabtagene Autoleucel Versus Rituximab Treatment in Participants With Severe Refractory Diffuse Cutaneous Systemic Sclerosis

12. Clinical Framework (Systemic Sclerosis Specific)

Primary Condition / Disease Areas: Scleroderma, Diffuse (Severe Refractory Diffuse Cutaneous Systemic Sclerosis [dcSSc]) **Diagnosis Threshold:** 2013 ACR/EULAR classification, dcSSc subset according to LeRoy, disease onset ≤ 7 years **Medical Methodology:** Anti-CD19 Chimeric Antigen Receptor T-cell (CAR-T) Therapy vs. B-cell depleting antibody **Optimization Target:** Achievement of an rCRISS50 response encompassing skin fibrosis reduction, functional stabilization, and preservation of lung capacity.

13. Study Procedures & Methodology

Management Pathway: Specialized cellular therapy pathway (Leukapheresis → Lymphodepletion → CAR-T Infusion) **Education Protocol (HCP):** Site certification for CAR-T administration, CRS management, and blinded mRSS standardization **Education Protocol (Patient):** Cellular therapy expectations, infection precautions, and long-term reporting **Quality Audit Mechanism:** Independent central review of clinical assessments and strict blinding verification

14. Observation & Follow-Up Schedule

Total Duration: 5 Years (approx. 260 weeks) for main study + ongoing separate LTFU protocol **Onsite Visit Cadence:** Intensive monitoring in Weeks 1-4 post-infusion; Quarterly through Year 1; Annually through Year 5 **Remote Monitoring Frequency:** Intermittent safety follow-ups as per cellular therapy standard guidelines **Checklist Review Interval:** Every 12 weeks (Q12W) for patient-reported outcomes (e.g., HAQ-DI)

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					